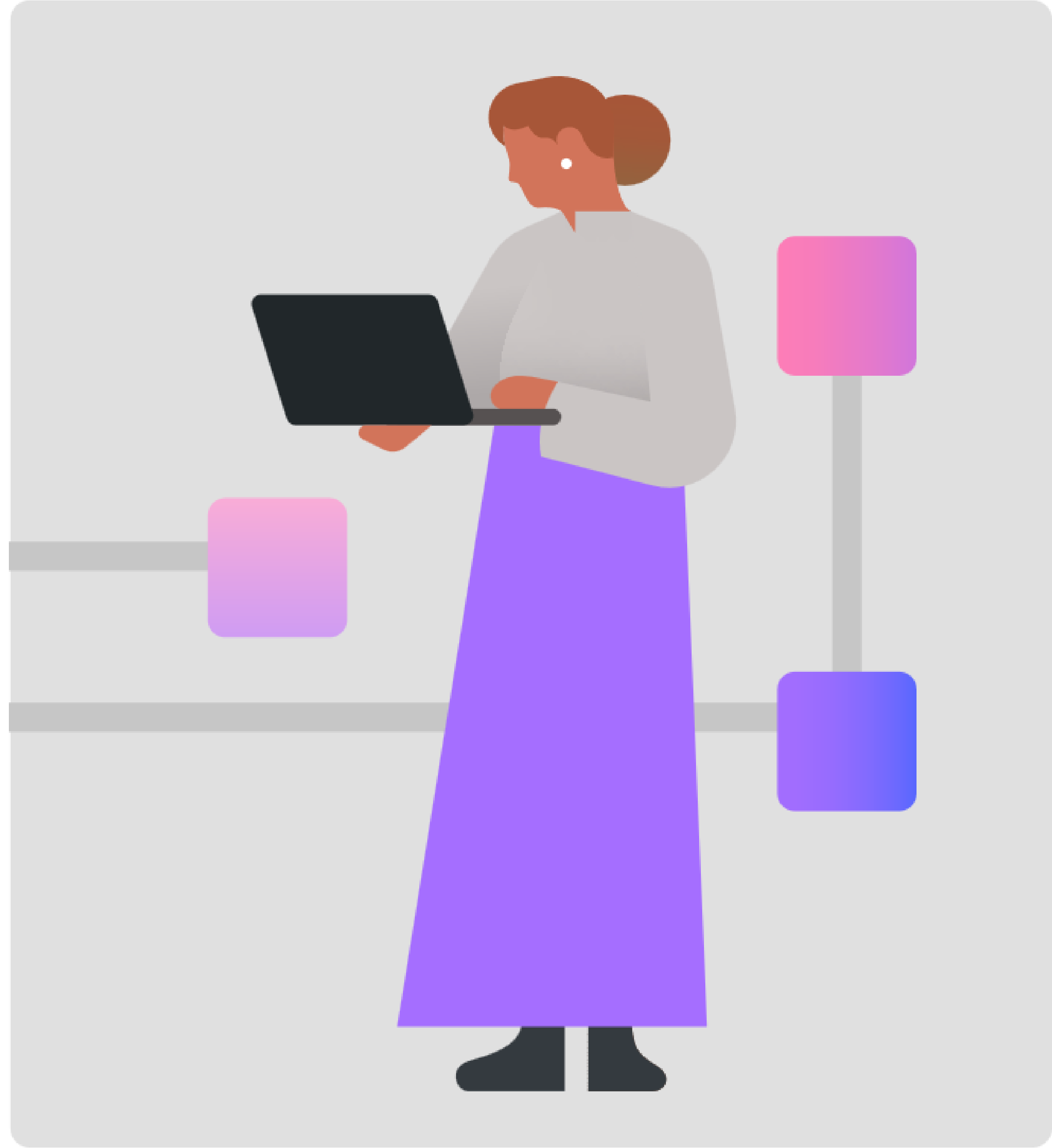
Qiskit Fall Fest 2025

Last Fall Fest IRL Meetup –

Intro to DiVincenzo Criteria + revisit previous questions

Natalie Hawkins
Qiskit Advocate, Tier 1
Seattle Quantum Computing Meetup, Founder



Agenda

Announcements

Intro to DiVincenzo Criteria

Q News – Industry & Academia

Revisit Previous Questions

Talk about the next Meetup

Mingle

Announcements

1

Submissions Deadline for earning a Winning Certificate is Extended to Monday, Nov 24!!!

The non-coding and Hackathon options can be starts on projects; they don't have to be fully worked out.

2

Project Idea: (1-2 pgs max total)
Short answers on questions:
- How will QC affect our futures?
- How will QC and AI affect our futures?
- What are the pro's and con's of each type of qubit?
- How many qubits are required for Shor's alg?

3

Our last in-person FF25 Meetup will be next Saturday, Nov 1, 2-4 pm, at the Montlake Branch of the SPL.

Topics TBD:
1) DiVincenzo Intermediate Notebook
2) Working in QC, Getting a job, Where are the jobs?

4

Upcoming Summit:

TechAlliance Policy Matters Summit | The Quantum Frontier

https://luma.com/sxwd8icd

TechAlliance Policy Matters Summit | The Quantum Frontier

Overview -

Washington has what it takes to lead in quantum top-tier research, major tech players, and a growing innovation ecosystem. What's missing is a coordinated strategy. The 2025 Policy Matters Summit will bring together policymakers, educators, industry, and startups to launch a shared vision for Washington's quantum future.

Summit Goals -

Position Washington as a hub for quantum R&D, talent, and commercialization
Spark collaboration and follow-on action statewide

Why Now -

States like Texas, Illinois, and Maryland are moving fast - passing laws, building hubs, and attracting funding. Washington has the assets. This Summit is our chance to catch up - and take the lead.

Summit Agenda

Wed, Nov 19, 2025

8:30 Breakfast/Networking

9:00 Opening remarks by Lieutenant Governor Denny Heck

9:20 **Quantum Basics** - A beginner-friendly overview of quantum concepts, technologies, and real-world applications

11:00 **Q-Day 101** - What is Q-Day and why does it matter? Learn how quantum will impact cybersecurity, infrastructure, and the economy.

12:00 Networking lunch

1:00 **The Quantum Landscape** – **National and Global Initiatives** - Explore how other states and countries around the world are advancing quantum

2:00 **Building a Quantum Ecosystem in WA**

Others Are Moving Fast

- passing laws
- building hubs
- attracting funding

Look at IEEE Quantum Week Locations:

- 2022 Broomfield, CO
- 2023 Seattle
- 2024 Montreal
- 2025 Albuquerque
- 2026 Toronto

https://qce.quantum.ieee.org/2025/

IL

Chicago Quantum Summit, Nov 3-4

https://chicagoquantum.org/summit
2025

MD

In 2025, Maryland's quantum ecosystem is experiencing significant growth, fueled by major investments, partnerships with technology leaders like Microsoft and DARPA, and the continued expansion of homegrown companies such as IonQ. (Google)

TX

Texas is solidifying its position as a major hub for quantum technology through the recently passed Texas Quantum Initiative, which aims to boost state investment and coordinate research and development. The effort involves collaborations between academic institutions, industry leaders, and state government. (Google) (**Abilene, Texas:** The first and flagship site is a massive AI data center $500B Stargate Project.)

What is Q-Day?

The quantum computing meaning of Q-Day

•**What it is:**
Q-Day is the day a powerful enough quantum computer can use algorithms like Shor's algorithm to crack asymmetric encryption that protects data like online banking, email, and other secure communications.

•**Why it's a concern:**
It would create a "cryptographic apocalypse," where sensitive data could be easily decrypted by attackers, impacting everything from financial transactions to secure messaging.

•**Current status:**
Experts debate the exact timeline, but they agree that organizations need to prepare by transitioning to post-quantum cryptography (PQC) to stay secure.

•**Preparing for Q-Day:**
The race is on to develop and implement new quantum-resistant encryption methods before a powerful quantum computer arrives.

The historical meaning of Q-Day

•**What it is:** In 1945, "Q-Day" was the name of the dress rehearsal for the Trinity atomic bomb test, the first detonation of a nuclear weapon.

5 Misconceptions about Q-Day
By Werner Coomans
24 May 2025

https://www.nokia.com/quantum/5-misconceptions-about-q-day/#:~:text=There%20are%20two%20families%20of,which%20are%20currently%20undergoing%20standardization.

Q News

« Darkness over America | The QMA Singularity »

HSBC unleashes yet another "qombie": a zombie claim of quantum advantage that isn't

Today, I got email after email asking me to comment on a new paper from HSBC—yes, the bank—together with IBM. The paper claims to use a quantum computer to get a 34% advantage in predictions of financial trading data. (See also blog posts here and here, or numerous popular articles that you can easily find and I won't link.) What have we got? Let's read the abstract:

Claims

"When and how will quantum computing broadly benefit humanity? Despite exhilarating recent progress, we still don't know. Here my friend Jens Eisert and I assess the current status and the challenges ahead. We are optimistic about the quantum future, but there's a lot of work to do."
John Preskill, Caltech.

https://lnkd.in/gxuzxhj3
2510.19928

What's Needed

"Gaps that we still have to cross in order to reach broadly useful technology. An important take-away: **the biggest gap is not in hardware**, **but in developing the right quantum algorithms**."
Koen Groenland, QuSoft.

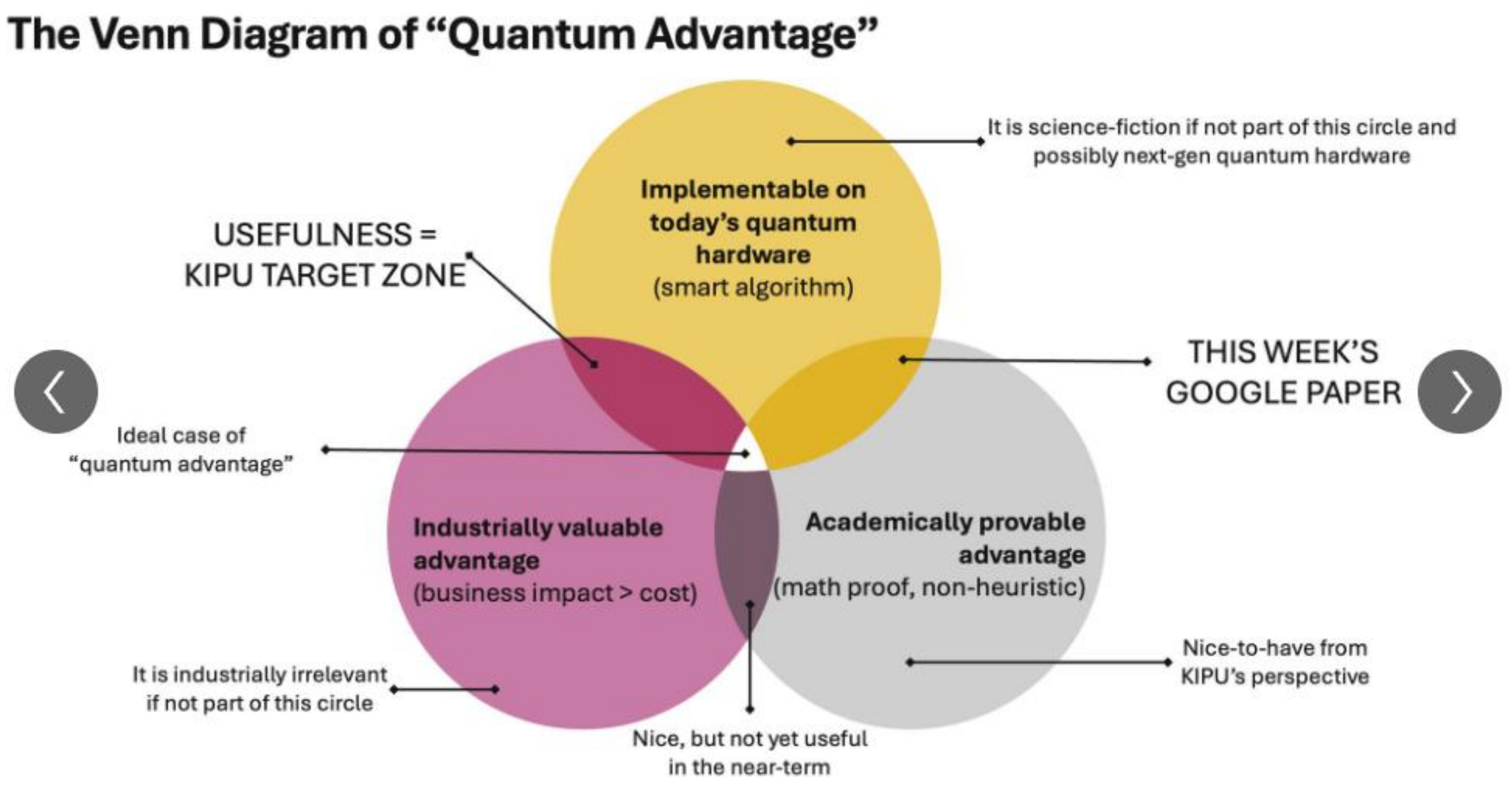
Mind the gaps: The fraught road to quantum advantage

Jens Eisert[1,2,3] and John Preskill[4,5]
[1]*Dahlem Center for Complex Quantum Systems, Freie Universität Berlin, 14195 Berlin, Germany*
[2]*Helmholtz-Zentrum Berlin für Materialien und Energie, 14109 Berlin, Germany*
[3]*Fraunhofer Heinrich Hertz Institute, 10587 Berlin, Germany*
[4]*Institute for Quantum Information and Matter, California Institute of Technology, CA 91125, USA*
[5]*AWS Center for Quantum Computing, Pasadena, CA 91125, USA*

QISKIT 2025 FALL FEST 2025

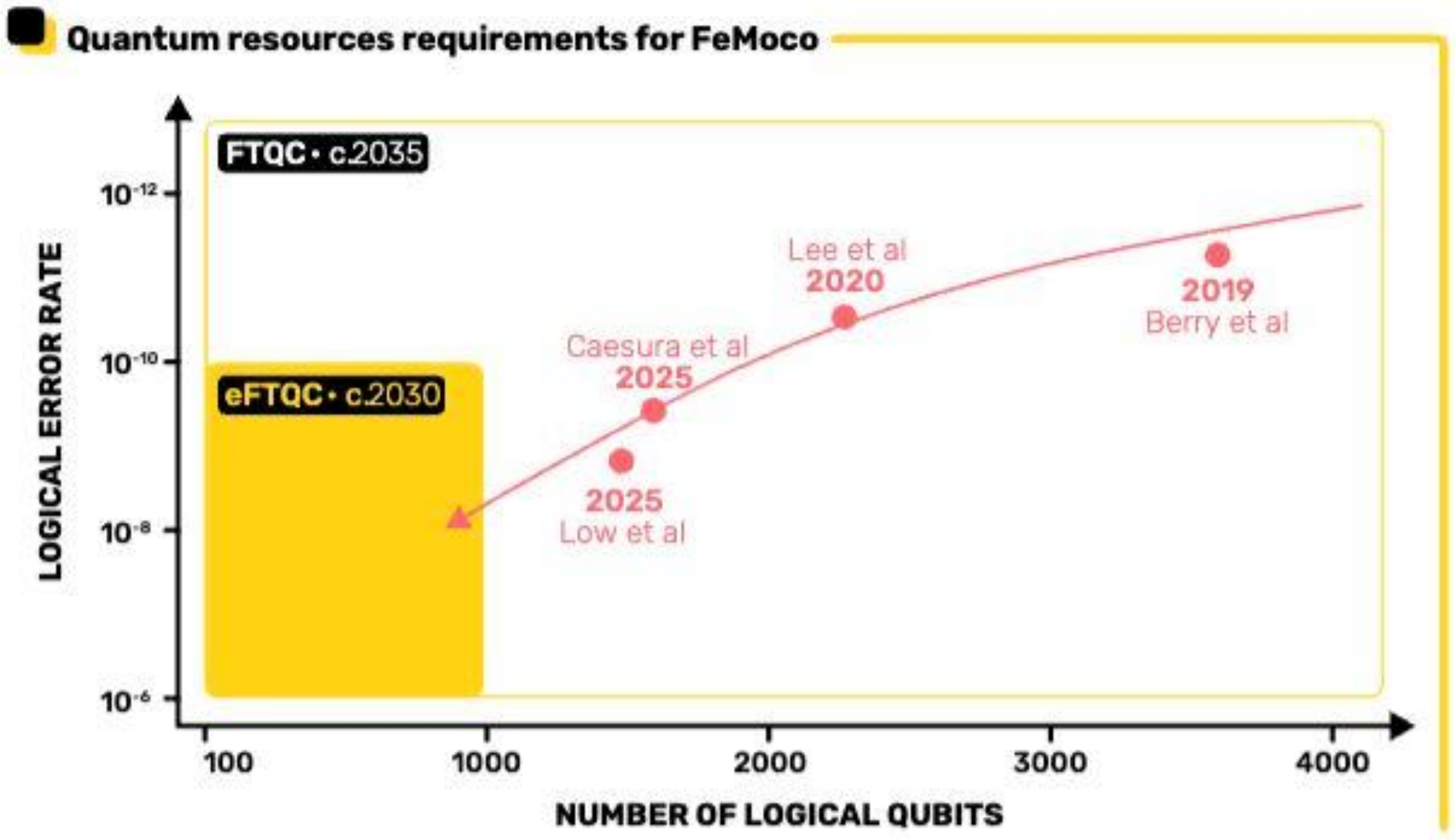
Q News

Sweet Spot for Q Advantage



Error Rates Needed

Koen Groenland, QuSoft, Amsterdam



Q News

Qubit Technology Comparison

Table compiled by Automatski, in India

Feature	Automatski	IBM	Google	Microsoft	IonQ	Rigetti	Atom Computing	PsiQuantum	Quantinuum
Qubit Technology	Photonic	Superconducting	Superconducting	Topological	Trapped Ions	Superconducting	Neutral Atoms	Photonic	Trapped Ion
#Qubits (Physical)	Undisclosed. Only focus on Logical Qubits	127 (eagle) 433 (Osprey) 1,121 (Condor) 133 (Heron)	53 (Sycamore) 105 (Willow)	Claim 8 (Disputed)	20 (Aria) 32 (Forte)	84 (Ankaa-3)	1,180	???	56
#Qubits (Logical)	5000+ (Available For Trials)	??? Refuses to Publish Benchmarks	??? Maybe 2	N/A	???	???	24	???	23
Gate Depth	50,000	5,000	???	N/A	800	???	???	???	4900
Encoding	Discrete	Discrete	Discrete	Discrete	Discrete	Discrete	Discrete	Continuous Variable	Discrete
Temperature	Cryogenics (Photon Detectors)	Cryogenics	Cryogenics	Cryogenics	Room temperature + laser cooling to µK	Cryogenics	Room temperature + laser cooling to µK	Cryogenics (Photon Detectors)	Room temperature + laser cooling to µK
Coherence	> 1 hour **extremely long coherence** since photons don't decohere thermally	300 µs	100 µs	N/A	> 1 hour	20 µs	> 40 secs	10's of secs **extremely long coherence** since photons don't decohere thermally	Several Minutes
Fidelity	99.999%	99.9%	99.9%	N/A	99.9%	99.9%	99.6%	99%	99.99%
Year Of Incorporation	1993	1911	1998	1975	2015	2013	2018	2016	2021
Year Of Launch	1997	2016	2019	Not Yet Launched. Just an Unproven Concept.	2018	2017	2021	Not Yet Launched	2021

DiVincenzo Criteria

Qiskit Coding Challenge Intermediate Notebook

Physicist David DiVincenzo outlined five key requirements for any physical implementation of a quantum computer, plus two additional criteria for quantum communication. In this notebook, we will experience each DiVincenzo criterion through practical Qiskit demonstrations. Rather than going deep into theory, each section briefly explains one criterion and then provides code exercises using Qiskit 2. You will get to run circuits on simulators and real IBM Quantum devices to explore each principle hands-on.

Five Criteria for Quantum Computation:

1. **A scalable physical system with well-characterized qubits.**

2. **Ability to initialize qubits** to a simple reference state (e.g. |00...0⟩).

3. **Long decoherence times** (qubit coherence much longer than gate operation time).

4. **A universal set of quantum gates** (able to perform arbitrary unitary operations).

5. **Qubit-specific measurement capability** (read out the state of each qubit).

(DiVincenzo also described two criteria for quantum communication: the ability to interconvert stationary and "flying" qubits, and to faithfully transmit flying qubits between locations. We include these in a recommended activity at the end of this notebook.)

Qiskit Global Summer School (QGSS25) Lecture by David DiVincenzo

The Physical Realization of Qubits (~ an hour)

https://www.youtube.com/watch?v=H4VnP1uL-eU

Revisit Previous Questions

1

Greek Alphabet

2

Data and Quantum Computers

- Data Loading Project
- Watch out for Bad Articles, written by AI or humans
- Small Data, Big Solution Space (Medium: J Krupansky)
- Now-ish vs. Later-ish (FTQC, qRAM)

3

AI + Quantum, Quantum + AI

- Qiskit Code Assistant
- AI Software Survey (TensorFlow?)
- Quantum Software Survey (TensorFlow Quantum?)

4

Computer Chips

- How SC qubits work, Qiskit Channel, Zlatko's talk, Parts 1 and 2, Lectures 16-21, (~6 hrs)
- How QPUs and CPUs are made

Greek Alphabet

Phi and Psi are used a lot, also Alpha, Beta, Pi, and Theta, and, well, the others too!

https://en.wikipedia.org/wiki/Greek_alphabet

AI and Quantum Computers

Qiskit Code Assistant

- IBM Quantum has one
- Available via a Premium Plan
- Designed to create Qiskit 2 code
- https://quantum.cloud.ibm.com/docs/en/guides/qiskit-code-assistant

AI Software Survey

- O'Reilly Tech Trends Report (from Jan 6 '25: based on '24, '23)
- Look for TensorFlow

- https://www.oreilly.com/pub/pr/3465

Quantum Open-Source Software Survey

- See the Unitary Foundation's annual survey

https://unitaryfoundation.github.io/survey-2024/

- Look for TensorFlow Quantum in the results
- Look for CUDA-Q

AI + Q, Q + AI

- AI helping Quantum
- Quantum helping AI
- Conference Q+AI
- Company: e.g. Sandbox AQ

- IEEE Quantum Week 2025, *can still register for on-demand content (~$200)*

- https://web.cvent.com/event/3f8c3754-761c-4db5-a3f5-20003fcc785c/summary

AI Tool Trends

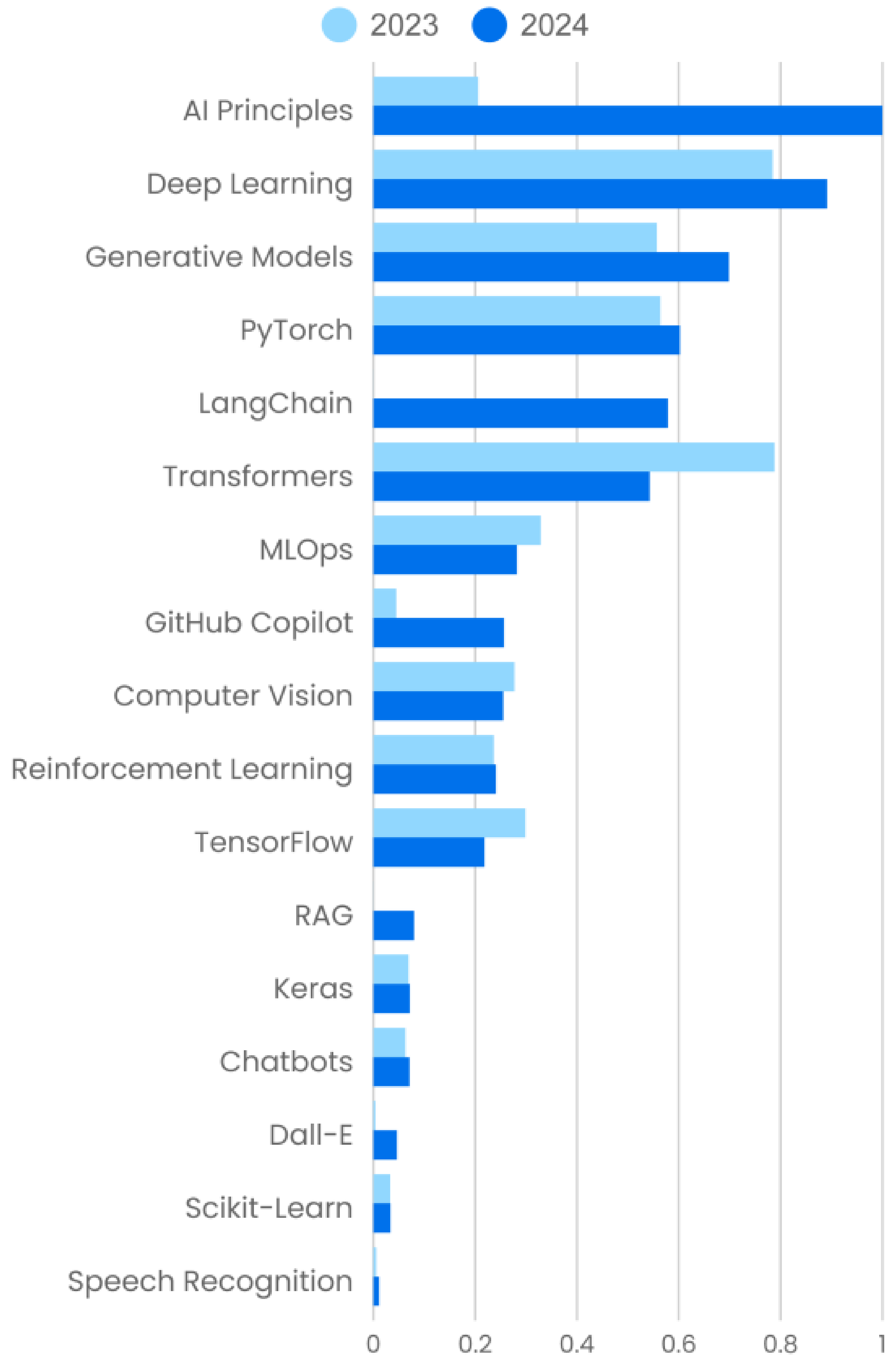


Figure 2. Skills needed for AI

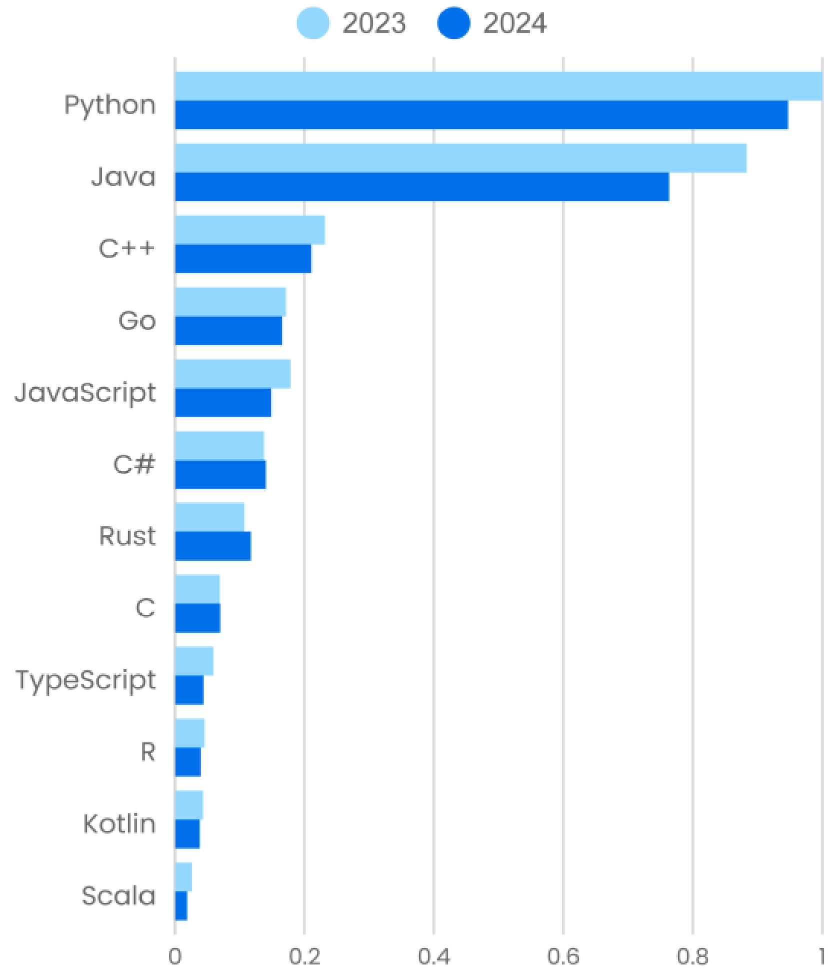


Figure 5. Programming languages

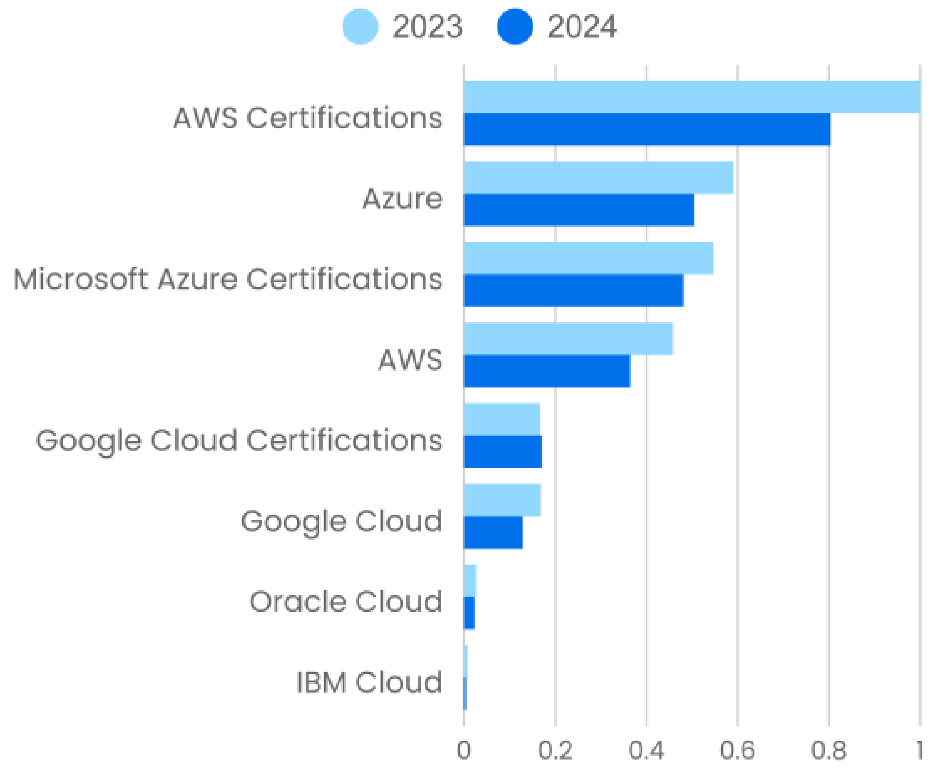


Figure 9. Cloud providers and certifications

Technology Trends for 2025 Report by O'Reilly, Jan 6 2025

https://www.niso.org/niso-io/2025/01/oreilly-offers-insights-tech-trends-2025#:~:text=Usage%20of%20material%20about%20Software,a%20bearing%20on%20software%20architecture.

How to Choose Between TensorFlow vs PyTorch in 2025

"**PyTorch is** easier to **use** and better for research, while **TensorFlow is** better for production and scalability."

http://youtube.com/watch?v=yOGi4vmtNaY

Panorama of the Open-Source Large Model Ecosystem 2.0

In Ant Group's newly released *Panorama of the Open-Source Large Model Ecosystem 2.0,* TensorFlow has been officially removed.

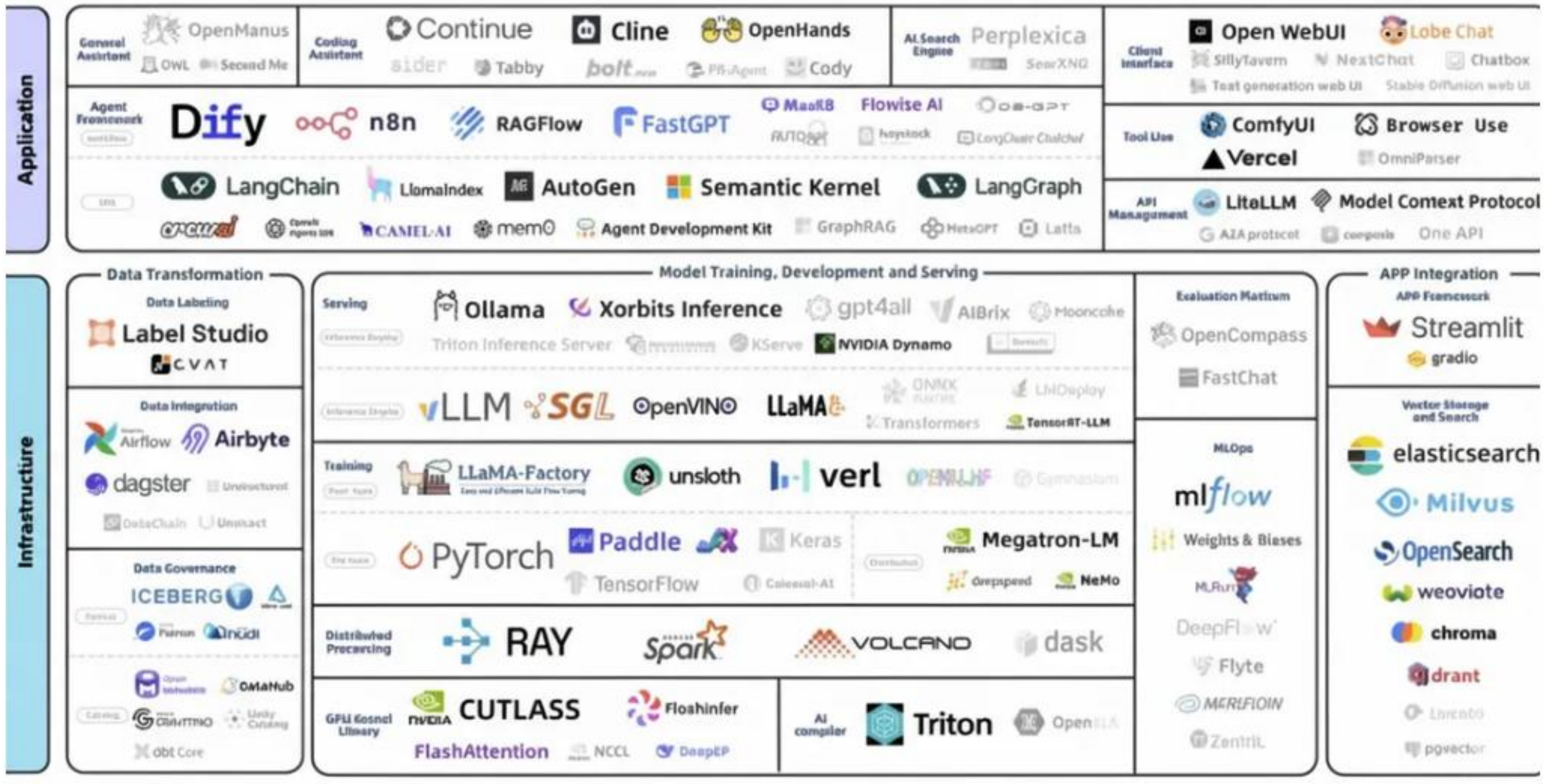
Medium article, *TensorFlow Is Dead. PyTorch Won.*, [Pan Xinghan](), Sep 15, 2025

How Industry: Cracks in TensorFlow's Fortress

"In industry, TensorFlow still hangs on, thanks to its early enterprise adoption. Big companies with legacy models — think Google Translate, some cloud pipelines, some banks — still run it. But the cool kids aren't choosing it anymore.

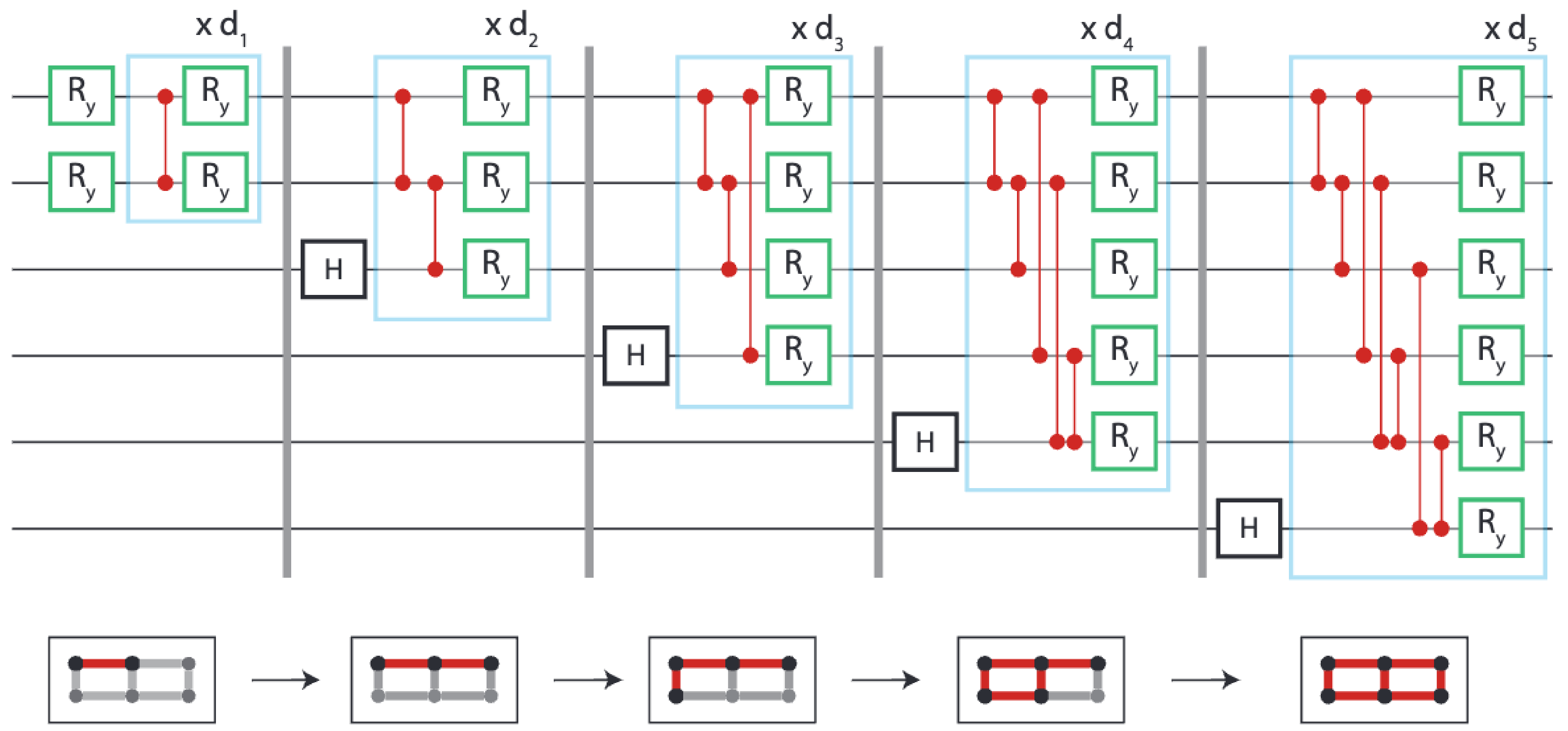
OpenAI trained GPT-3 and DALL·E on PyTorch. Tesla's Autopilot runs on PyTorch. Meta, Microsoft, and Amazon have rallied behind it. NVIDIA and Intel are optimizing PyTorch first. Even Google Cloud now openly supports PyTorch and quietly shifts research toward JAX instead of TensorFlow.

Yes, TensorFlow has stars on GitHub. But stars don't equal momentum. Forks, contributors, and new repos overwhelmingly lean PyTorch. The trend lines are clear: TensorFlow is legacy, PyTorch is future."



Panorama of the Open-Source Large Model Ecosystem 2.0 released by Ant Group

Data and Quantum Computers



Blue Qubit, 2311.12929 https://www.bluequbit.io/quantum-data-loading

Figure 1: A circuit ansatz with hierarchical architecture on 6 qubits laid out on a grid. Vertical gray lines on the circuit indicate a partial circuit which is variationally learned before using it as input to the next sub-circuit.

Quantum State Preparation

Phys. Rev. Lett. **129**, 230504 – **Published 30 November, 2022**

"Quantum state preparation is an important subroutine for quantum computing. We show that any n-qubit quantum state can be prepared with a $\Theta(n)$-depth circuit using only single- and two-qubit gates, although with a cost of an exponential amount of ancillary qubits."

Little Data with Big Solution Space

"No, quantum computers are not appropriate for *big data* problems. Rather, they are best for problems with a fairly small amount of data which has a very large solution space — so called *combinatorial explosions*. So, rather than call it *Big Data*, I call it *Little Data* with a *Big Solution Space*. This informal paper introduces the notions of *Little Data with a Big Solution Space*."

https://jackkrupansky.medium.com/little-data-with-a-big-solution-space-the-sweet-spot-for-quantum-computing-b6c6d76ada6d

Quantum Resource Estimation for Data Loading or State Preparation in Quantum Computers

Google AI Overview

"Resource estimation for data loading and state preparation is a critical aspect of analyzing the overall cost of a quantum algorithm. The resources required often scale exponentially with the number of qubits for preparing a generic quantum state. However, tailored approaches can offer substantial reductions in circuit depth and qubit requirements for specific problems.

Key resource metrics include the number of qubits, quantum gates, and execution time. This process is particularly challenging because it involves translating classical data into a quantum format.

Encoded classical data: In quantum machine learning (QML) and other data-driven algorithms, classical data must be loaded into a quantum state. The efficiency of this process is a major bottleneck."

Quantum Applications

Data and Quantum Computers

Qiskit Functions

https://quantum.cloud.ibm.com/functions

Multiverse Computing

Singularity Machine Learning Function

“With the Singularity Machine Learning function, you can solve real-world classification problems on quantum hardware without requiring quantum expertise. This function, based on ensemble methods, is a hybrid classifier which combines classical methods like boosting for ensemble training with quantum methods like quantum approximate optimization (QAOA) for optimizing the trained ensemble for maximum diversity, and generalization. Unlike other quantum machine learning solutions, this function is capable of handling large-scale datasets with millions of examples and features without being limited by the number of qubits. It is also highly flexible, and can be used to solve classification problems across a wide range of domains, including finance, healthcare, and cybersecurity.”

Quantum Resource Estimation

Quantum Computing Report, article, *Update on the GQI Quantum Resource Estimator Playbook*

https://quantumcomputingreport.com/update-on-the-gqi-quantum-resource-estimator-playbook/

Resource Estimation Tools, Google AI Overview

Tools for quantum resource estimation include **Azure Quantum Resource Estimator**, **QREChem**, **TopQAD**, and **PsiQuantum's QREF and Bartiq**. These tools help estimate the logical and physical qubits, runtime, and other resources needed for a quantum program on a fault-tolerant quantum computer by taking into account various parameters like hardware specifications and error correction codes.

What is QRE and why is it hard?

https://www.mustythoughts.com/perfect-qre-sw

Superconducting Qubits

Superconducting Qubits I: Quantizing a Harmonic Oscillator, Josephson Junctions - Part 1, Lecture 16, by Zlatko Minev, IBM Quantum (now at Google)

https://www.youtube.com/watch?v=kV0RsKXSqRg

Also, see Lectures 17-21, approx. 6 hours in total

Zlatko has several presentations on noise too. One example,

https://www.zlatko-minev.com/blog/quantum-error-mitigation-bss23

How are QPUs Fabricated?

Inside Quantum Chip Manufacturing: From Qubits to Fabrication, SpinQ Blog

https://www.spinquanta.com/news-detail/inside-quantum-chip-manufacturing-from-qubits-to-fabrication

How To Build A Quantum Computer

YT Channel: Lukas's Lab (9 min)
https://www.youtube.com/watch?v=N06hC1GL1ns&t=421s

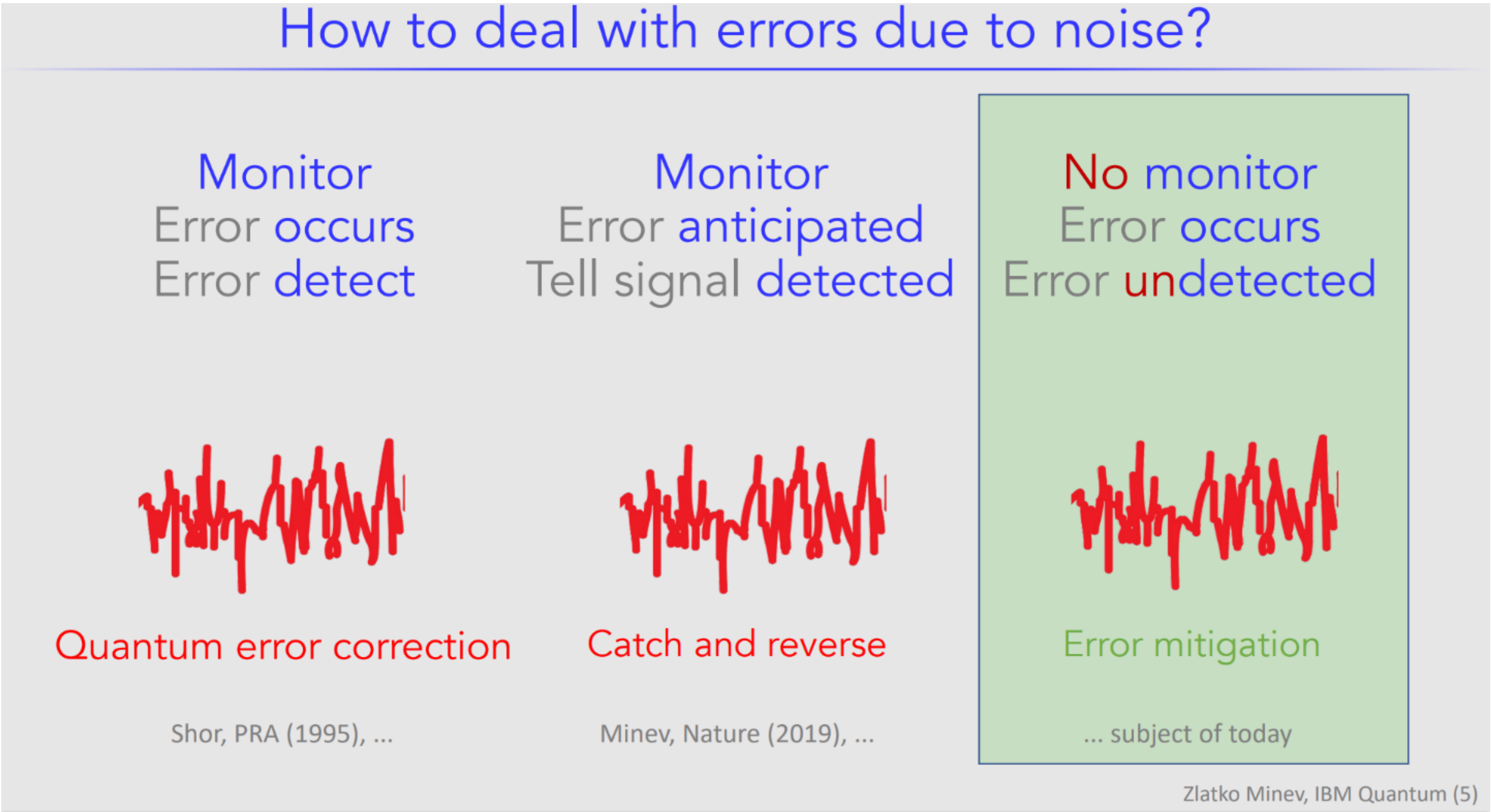
Intel's Big Bet: Inside the Chipmaker's Make-or-Break Factory

https://www.nytimes.com/2025/10/09/technology/intel-chip-factory-fab52.html?searchResultPosition=1

How Semiconductors Are Fabricated

https://www.nytimes.com/2022/04/08/technology/intel-chip-shortage.html

QC Biggest Challenge?

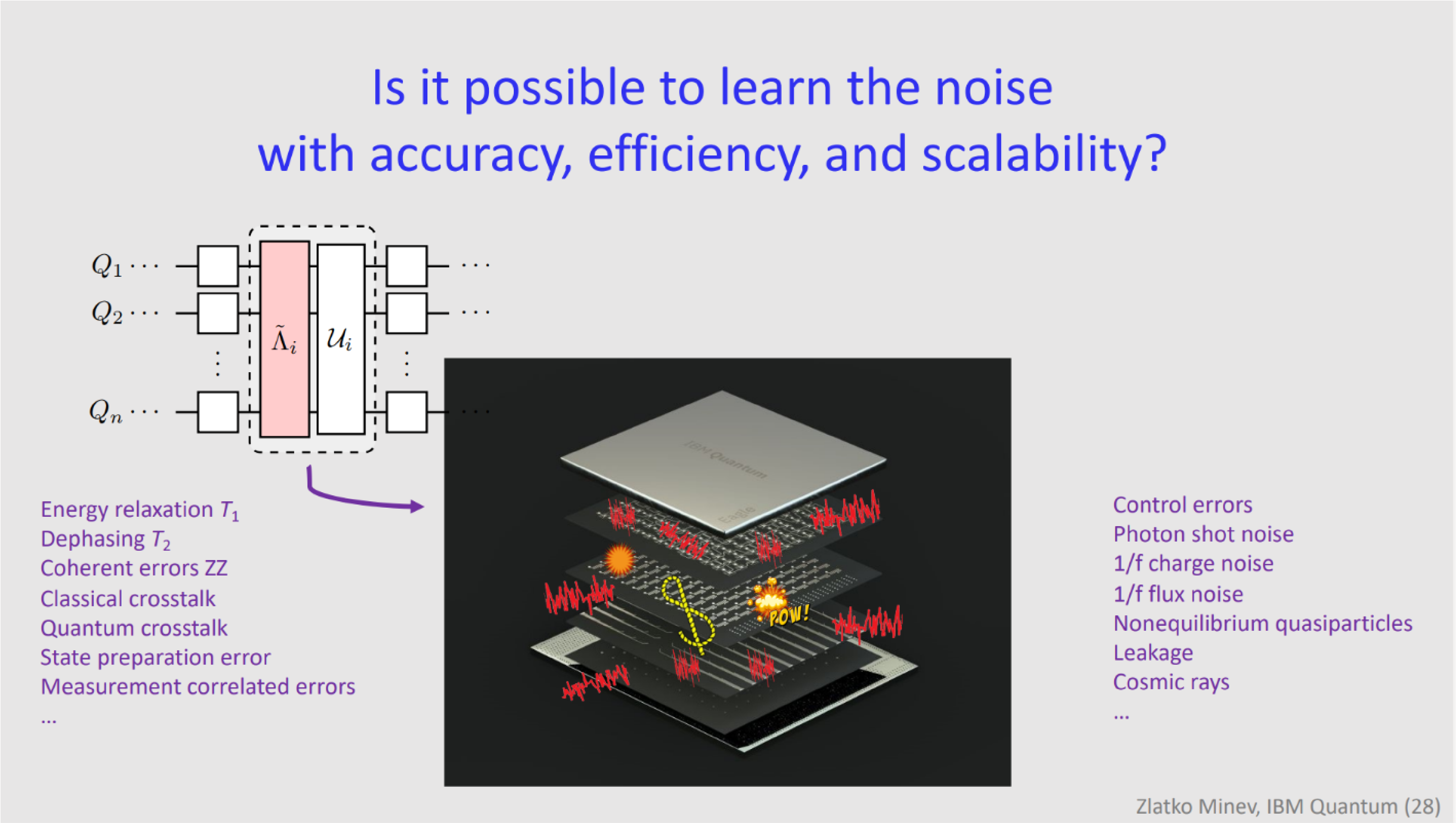


Zlatko says, “Noise.” (2022)

https://www.zlatko-minev.com/blog/pec-talk

How Deal with Noise?

Depends on the type of noise.



Qiskit Beginner Challenge Notebook

1

Software Setup

2

Circuits and Visualization

3

Optimize, then Run on Simulator or Fake Backend

4

Run on Real Quantum HW

Qiskit Fall Fest 2025

Thanks for Coming!!!

Deadline for Submissions: Monday Nov 24, 10 am PT

Upcoming Event: TechAlliance, Nov 19, South Seattle

Upcoming Meetup: Date? in December w/Holiday Party